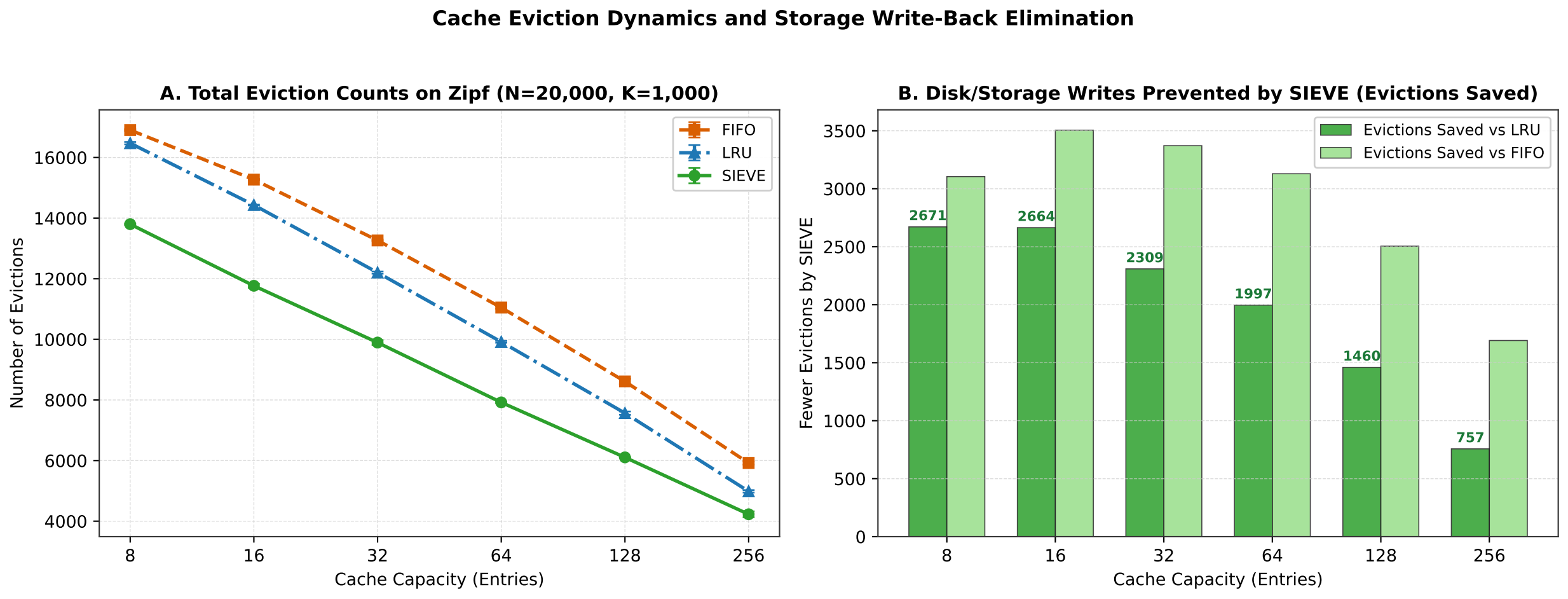
Cache Eviction Dynamics and Storage Write-Back Elimination
A. Total Eviction Counts on Zipf (N=20,000, K=1,000)
FIFO
LRU
SIEVE
Number of Evictions
16000
14000
12000
10000
8000
6000
4000
8
16
32
64
128
256
Cache Capacity (Entries)
B. Disk/Storage Writes Prevented by SIEVE (Evictions Saved)
Evictions Saved vs LRU
Evictions Saved vs FIFO
Fewer Evictions by SIEVE
3500
3000
2500
2000
1500
1000
500
0
2671
2664
2309
1997
1460
757
8
16
32
64
128
256
Cache Capacity (Entries)